CLOUD FUNDAMENTALS – AWS

1. Objective

The objective of this assignment was to gain practical knowledge of fundamental AWS cloud services by configuring compute, storage, and access-management resources.

The following activities were performed:

- Launching and configuring an Amazon EC2 instance.
- Creating an Amazon S3 bucket with lifecycle management.
- Creating an IAM role with least-privilege permissions.
- Configuring an EC2 Auto Scaling Group.
- Verifying the AWS resources through the AWS Management Console.

2. AWS Services Used

AWS Service	Purpose
Amazon EC2	Provides virtual compute capacity for running applications.
Amazon S3	Provides scalable object storage.
AWS IAM	Manages roles, permissions, and access to AWS resources.
EC2 Auto Scaling	Maintains the required number of EC2 instances.

3. EC2 Instance – Launch and Configuration

Amazon EC2 (Elastic Compute Cloud) is a cloud service that provides resizable virtual servers. EC2 instances can be configured with different instance types, operating systems, networking settings, storage, and security configurations.

An EC2 instance was launched as part of the assignment and verified through the AWS Management Console.

EC2 Configuration

Parameter	Value
Instance Name	`test2`
Instance ID	`i-0d7b40898bcda0231`
Instance Type	`t3.micro`
Instance State	Running

AWS Region	US West (Oregon)
Public IPv4	`44.251.231.31`
Private IPv4	`172.31.28.138`
IPv6	Not assigned
Elastic IP	Not assigned

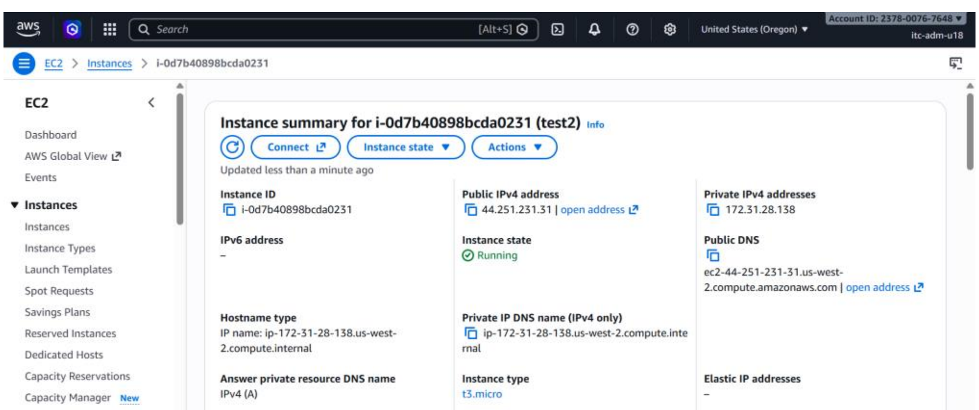
Procedure

1. Open the AWS Management Console.
2. Navigate to **EC2 → Instances**.
3. Select **Launch Instance**.
4. Provide the instance name.
5. Select the required Amazon Machine Image.
6. Select the `t3.micro` instance type.
7. Configure the required key pair, network, security group, and storage settings.
8. Launch the instance.
9. Open the Instances page and verify the instance state.
10. Confirm that the instance reaches the **Running** state.

Verification

The EC2 console confirmed that instance `i-0d7b40898bcda0231`, named `test2`, was successfully launched. The instance was in the **Running** state and was configured using the `t3.micro` instance type.

Figure 1: EC2 Instance Configuration



4. S3 Bucket and Lifecycle Policy

Amazon S3 (Simple Storage Service) is an object storage service used to store and retrieve data. Data is stored as objects within S3 buckets.

For this assignment, the following S3 bucket was created:

Bucket Name: `itc-adm-u18-0209`

The AWS console shows that the bucket was successfully created and contained zero objects at the time of verification.

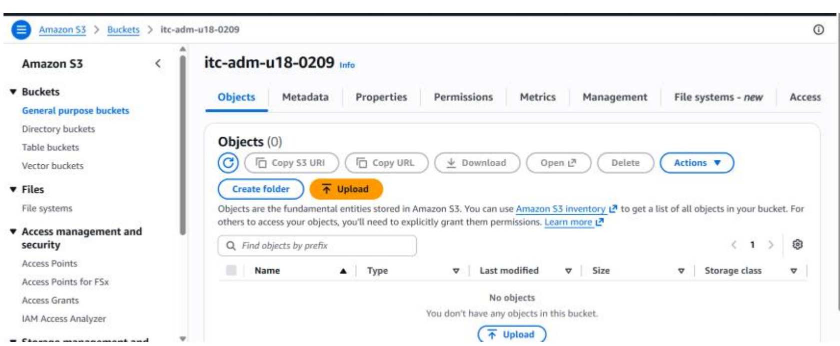
S3 Configuration

Parameter	Value
Bucket Name	`itc-adm-u18-0209`
Service	Amazon S3
Objects	0 at time of verification
Lifecycle Management	Configured

Procedure

1. Open the AWS Management Console.
2. Navigate to **Amazon S3**.
3. Select **Buckets**.
4. Select **Create bucket**.
5. Enter the bucket name `itc-adm-u18-0209`.
6. Configure the required region and security settings.
7. Create the bucket.
8. Open the bucket and verify that it is available.

Figure 2: S3 Bucket



4.1 S3 Lifecycle Policy

An S3 Lifecycle policy allows objects to be automatically transitioned or deleted according to defined rules. Lifecycle management can reduce storage costs and automate the retention of objects.

The lifecycle configuration is accessed through:

S3 → Bucket → Management → Lifecycle rules

A lifecycle rule was configured for the bucket to automatically manage objects according to the defined retention period.

Lifecycle Policy

```
{
  "Rules": [
    {
      "ID": "ExpireObjectsAfter30Days",
      "Status": "Enabled",
      "Filter": {
        "Prefix": ""
      },
      "Expiration": {
        "Days": 30
      }
    }
  ]
}
```

This configuration automatically expires objects after 30 days.

The lifecycle policy helps automate object management and avoids unnecessary long-term storage of objects that are no longer required.

5. IAM Role and Least-Privilege Policy

AWS Identity and Access Management (IAM) is used to control access to AWS services and resources. IAM roles allow AWS services such as EC2 to obtain permissions without storing long-term credentials on the instance.

An IAM role named **`adm-annapoorna`** was created for EC2.

IAM Role Configuration

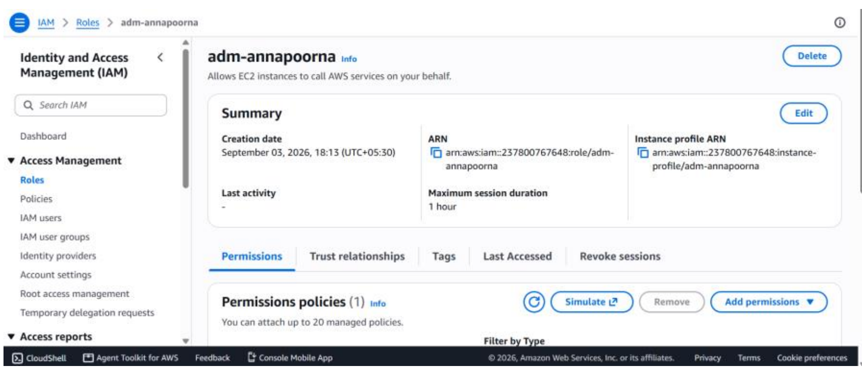
Parameter	Value
Role Name	`adm-annapoorna`
Purpose	Allows EC2 instances to call AWS services
Instance Profile	`adm-annapoorna`

Maximum Session Duration	1 hour
Permission Policies	1
Trusted Service	EC2

Procedure

1. Open the AWS Management Console.
2. Navigate to **IAM → Roles**.
3. Select **Create role**.
4. Select **EC2** as the trusted entity.
5. Attach the required permissions policy.
6. Enter the role name `adm-annapoorna`.
7. Review the configuration.
8. Create the role.
9. Verify the role and associated instance profile.

Figure 3: IAM Role Configuration



5.1 Least-Privilege Policy

The principle of least privilege requires granting only the permissions necessary to perform a specific task.

For an EC2 workload requiring read access to the S3 bucket, the following policy restricts permissions to the required bucket:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowListSpecificBucket",
      "Effect": "Allow",
      "Action": [
```

```
        "s3:ListBucket"
      ],
      "Resource": "arn:aws:s3:::itc-adm-u18-0209"
    },
    {
      "Sid": "AllowReadObjectsFromSpecificBucket",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::itc-adm-u18-0209/*"
    }
  ]
}
```

The `s3:ListBucket` permission allows listing objects in the specified bucket, while `s3:GetObject` allows retrieving objects from the bucket.

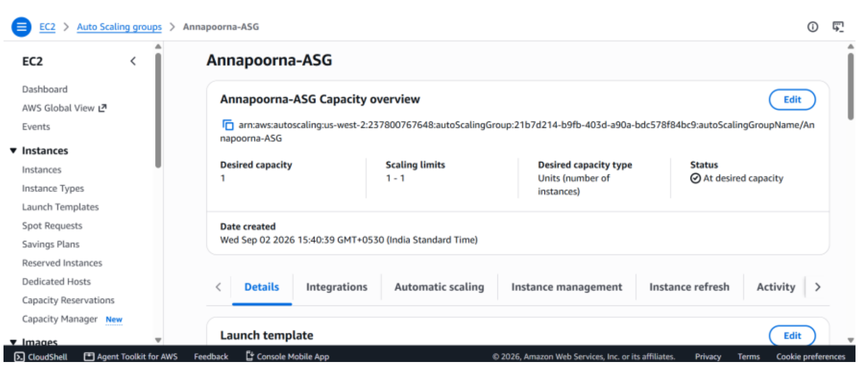
The policy does not provide permissions to delete objects, upload objects, create buckets, or administer other S3 resources. This limits the role's access and follows the principle of least privilege.

6. EC2 Auto Scaling Group

An EC2 Auto Scaling Group helps maintain the required number of EC2 instances and can automatically adjust capacity according to configured scaling requirements.

An Auto Scaling Group named **`Annapoorna-ASG`** was configured.

Figure 4: Auto Scaling Group Configuration



7. Verification and Security

The configured resources were verified through the AWS Management Console.

The following results were confirmed:

- The EC2 instance `test2` was successfully launched and is in the **Running** state.
- The EC2 instance uses the `t3.micro` instance type.
- The S3 bucket `itc-adm-u18-0209` was successfully created.
- The S3 bucket was available for object storage.
- The IAM role `adm-annapoorna` was successfully created for EC2.
- An instance profile was associated with the IAM role.
- A least-privilege policy was configured to restrict S3 access.
- The `Annapoorna-ASG` Auto Scaling Group was successfully configured and is at its desired capacity.

Security was addressed by restricting IAM permissions to required resources and actions. Using an IAM role for EC2 avoids the need to store long-term AWS credentials on the instance. S3 lifecycle management also provides automated control over object retention.

8. Conclusion

The AWS Cloud Fundamentals assignment provided practical experience with core AWS services including **Amazon EC2, Amazon S3, AWS IAM, and EC2 Auto Scaling**.

An EC2 instance was successfully launched and verified in the Running state. An S3 bucket was created and configured for lifecycle management. An IAM role was created for EC2, with permissions designed according to the principle of least privilege. The Auto Scaling Group configuration was also verified.

This assignment provided hands-on understanding of AWS compute, storage, identity and access management, lifecycle policies, IAM roles and policies, and basic cloud security practices.

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